Time series forecasting

Generic supervised learning approach:

samples are i.i.d. → not connected to each other.

{ samples before time t.

→ sample at time t can use previous samples to understand the history and come up better prediction.

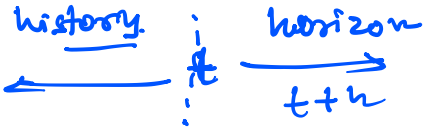
Bike rental dataset on kaggle.

Forecasting problem:

$$\hat{y}_{t+h|t} = f(y_t, y_{t-1}, \dots, x_t, x_{t-1|t^-})$$

h = horizon (how far ahead we predict)

t = forecast origin



Frequency: spacing b/w the samples.

Information set: Dataset (target and features)

Forecasting contract:

$$\hat{y}_{t+h|t} = f(I_t)$$

① Trend: T_t Sustained long-term increase/decrease.

② seasonality: S_t A pattern tied to known and fixed period m. can be weekly, monthly, yearly.

③ Cycle: A rise and fall without a fixed calendar period.

④ Remainder: residual R_t what remains after trend and seasonal pattern is removed.

$$y_t = T_t + S_t + R_t$$ Additive model.

$$y_t = T_t \times S_t \times R_t$$ multiplicative models

* **Baseline:**

① Mean:

$$\hat{y}_{t+h|t} = \bar{y}_t$$

② Naive:

$$\hat{y}_{t+h|t} = y_t$$

③ moving average:

$$\hat{y}_{t+1|t} = \frac{\overline{y_{t-1}} + y_t}{2}$$

$$= \frac{1}{w} \sum_{j=0}^{J} y_{t-j}$$

④ Drift:

$$\hat{y}_{t+h|t} = y_t + h \cdot \frac{y_t - y_1}{t-1}$$

Extends the avg. change from first to latest observation.

⑤ seasoning:

$$\hat{y}_{t+h|t} = y_{t+h-m(k+1)}$$

Repeats the matching position from the latest observation with season, $m(k+1)$

* Evaluate:

RMSE: $\sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$

MAPE: $\frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{|y|} \times 100$

$$\text{MAE}: \quad \sum_{i=1}^{n} |y_i - \hat{y}_i|$$

$$\text{RMSLE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} \left(\log(y_i+1) - \log(\hat{y}_i+1)\right)^2}$$

* **Lag:** Lag aligns today's target values that were already known.

Supervised framing.

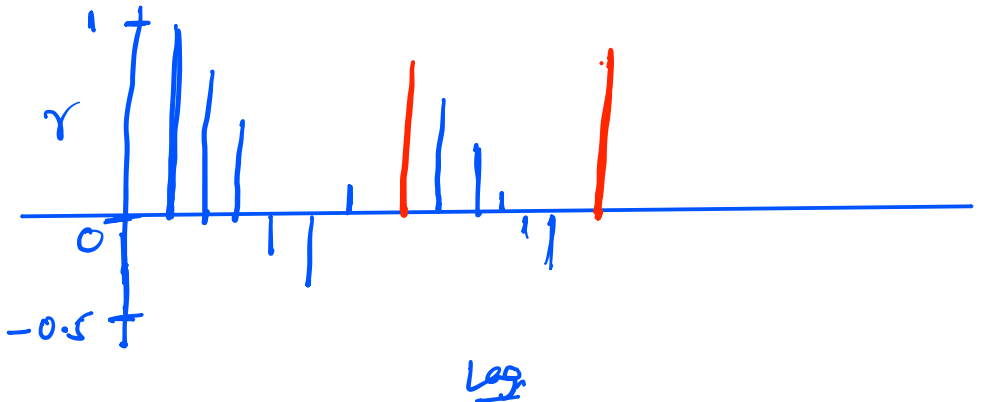
$x_t = [y_{t-1}, y_{t-2}, \ldots, y_{t-p}]$. target $= y_t$

Time	Lag 1	Lag 7	Last 7-day mean	target
Day t	y_{t-1}	y_{t-7}	$\frac{1}{7}\sum_{j=1}^{7} y_{t-j}$	y_t

ACF ⇒ Auto correlation function:

How strongly this series resembles delayed version of itself.

$$r_p = \frac{\sum_{t=p+1}^{T} (y_t - \bar{y})(y_{t-p} - \bar{y})}{\sum_{t=1}^{T} (y_t - \bar{y})^2}$$



Input → Hi how are you ? ?

GPT → most probable next word. → Hi

Input → Hi, how are you? Hi

* **Stationarity:**

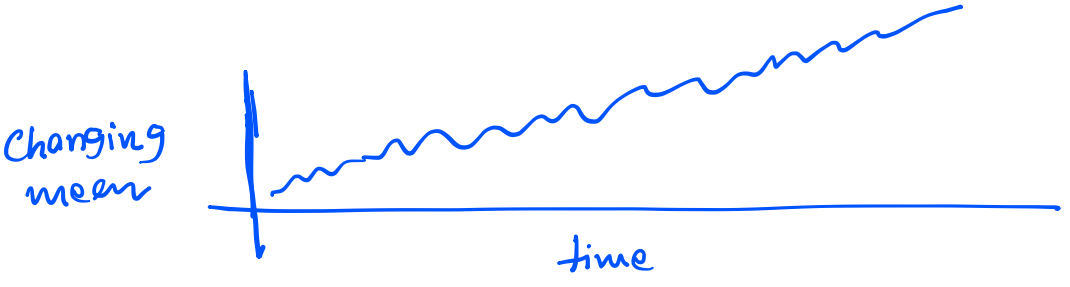
whether the statistical behaviour in one period remains a useful descriptor for another period.

Stationary looking data:

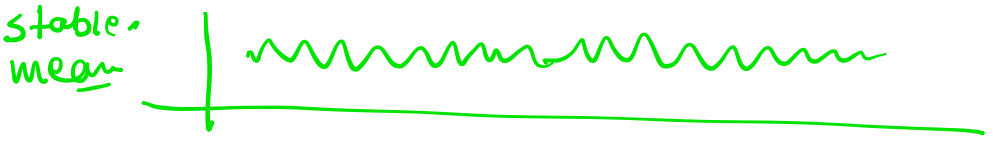
98, 103, 100, 96, 102, 101, 99 - - -

non-stationary looking data:

100, 108, 115, 130, 134, - - -







① Stable mean: $E(Y_t) = \mu$

② Stable variance: $var(Y_t) = \sigma^2$

Stationarity says its statistical behaviour is stable, not that its future values are known.

It helps: ① model generalization

② ACF are interpretable.

How to find stationarity:

① plot the data over time.

② compare time windows.

③ Augmented - Dickey - Fuller test
→ P value.

* Differencing:

$$\Delta Y_t = Y_t - Y_{t-1}$$

Y_t ↓ today, Y_{t-1} ↓ yesterday

100, 108, 116, 124, 132 - - - -

+8 +8 +8 - - -

First difference:

$$\Delta Y_t = Y_t - Y_{t-1} = (1 - B)Y_t$$

B → backshift operator

$$\boxed{BY_t = Y_{t-1}}$$

Seasonal difference:

$$\Delta_m Y_t = Y_t - Y_{t-m} = (1 - B^m)Y_t$$

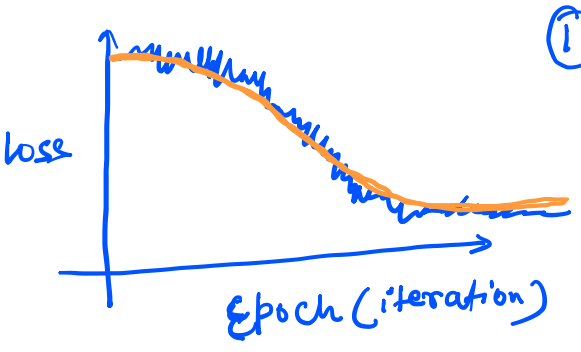
m = period.

Try to forecast changes, not the actual values.

differences

Smoothing: Tries to reduce the effect of short-term noise.

100, 104, 98, 103, 97, 105, 101



① moving average:

3-day MA:

$$M_t = \frac{Y_t + Y_{t-1} + Y_{t-2}}{3}$$

Reduces noise.

Limitation:

1. Fluctuations are hard to capture.
2. All points are equally weighted.

* Exponential Smoothing:

$$l_t = \alpha Y_t + (1-\alpha) l_{t-1}$$

(l_t: level)

Y_t = new observation

l_{t-1} = previous level estimate

α = smoothing parameter, $0 \le \alpha \le 1$

$l_0 = y_0$

$$l_1 = \alpha y_1 + (1-\alpha) l_0$$

$$l_2 = \alpha y_2 + (1-\alpha) l_1$$

(l_2: new belief; αy_2: portion of new evidence; $(1-\alpha) l_1$: portion of old belief.)

$l_{t-1} = 100$, $Y_t = 120$, $\alpha = 0.3$

$$l_t = 0.3 \times 120 + 0.7 \times 100$$

$$= 36 + 70$$

$$= 106$$

if $\alpha = 0.1$ → Trusts history more, reacts slowly, less sensitive to noise.

if $\alpha = 0.9$ → opposite of above.

why is it called Exponential:

$$l_t = \alpha Y_t + (1-\alpha) l_{t-1}$$

$$l_{t-1} = \alpha Y_{t-1} + (1-\alpha) l_{t-2}$$

$$l_t = \alpha Y_t + \alpha(1-\alpha) Y_{t-1} + \alpha(1-\alpha^2) Y_{t-2}$$

$$\alpha,\ \alpha(1-\alpha),\ \alpha(1-\alpha)^2,\ \text{-----}$$

Decline is exponential,

* forecasting using SES:

Yes we can, but trend and seasonality is missing,

Let's add trend,

l_t = current level

b_t = current trend

Holt's method:

$$l_t = \alpha Y_t + (1-\alpha)(l_{t-1} + b_{t-1})$$

$$b_t = \beta (l_t - l_{t-1}) + (1-\beta) b_{t-1}$$

$$b_0 = (Y_1 - Y_0)$$

$$\hat{Y}_{t+h|t} = l_t + h\, b_t$$

$0 \le \beta \le 1$

Holt's - Winter's method !

Add Seasonality !

l_t = level

b_t = trend

s_t = seasonal effect

Remove the known seasonal effect

$$l_t = \alpha(Y_t - s_{t-m}) + (1-\alpha)(l_{t-1} + b_{t-1})$$

$$s_t = \gamma(Y_t - l_{t-1} - b_{t-1}) + (1-\gamma)s_{t-m}$$

m = period

$$\hat{Y}_{t+h|t} = l_t + h \cdot b_t + s_t$$

ARIMA !

models a stationary version of the series using its past values and past forecast errors.

$$ARIMA(p, d, q)$$

AR → Auto Regressive → using past to predict the current

I → Integrated → difference of series.

MA → moving average → using prev. forecast error to predict the current

- p → no. of Autoregressive lags
- d → no. of differences
- q → no. of lagged Errors.

① AR part:

Auto → self

Regression → predicting values using past values

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots\cdots + \phi_p Y_{t-p} + \varepsilon_t$$

$p = 1$ (one lag)

$$Y_t = c + \phi_1 Y_{t-1} + \varepsilon_t \longrightarrow AR(1)$$

if ϕ is close to 1 → fluctuation persist for a long time

if ϕ " " " 0 → past has little influence

if $\phi < 0$ → Alternate around mean.

Desire → $\phi < 1$

② I part:

Random walk:

$$Y_t = Y_{t-1} + \varepsilon_t$$

$$\Delta Y_t = Y_t - Y_{t-1} = \varepsilon_t$$

① Diff. the series before modeling

② Forecast the differences.

$d = 0 \Rightarrow$ no difference,

$d = 1 \Rightarrow$ first difference

$d = 2 \Rightarrow$ second difference

ARC(1,1) $\Rightarrow$ 1 lag and 1 difference.

$$\Delta Y_t = Y_t - Y_{t-1}$$

$$\Delta^2 y_t = \Delta y_t - \Delta y_{t-1}$$

ARIMA model predicts :- $\hat{\Delta} Y_{t+1} = 8$

$$Y_t = 130$$

$$\hat{Y}_{t+1|t} = 130 + 8 = 138.$$

Part 3: MA :

MA model, uses prev. forecast errors.

$$Y_t = c + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

MA(1) $\Rightarrow$

$$Y_t = c + \varepsilon_t + \theta \varepsilon_{t-1}$$

ε_t is unknown but previous errors are known.

$$Y_t = \underbrace{100}_{c} + \underbrace{0.6}_{\theta}\,\varepsilon_{t-1} + \varepsilon_t$$

Let's assume $\varepsilon_{t-1} = 10$

$$\hat{Y}_t = 100 + 0.6(10)$$

$$= 100 + 6 = 106$$

Complete ARIMA model:

Differenced series:

$$Z_t = (1-B)^d Y_t$$

B is the backshift operator:

$$B Y_t = Y_{t-1}$$

$$Z_t = c + \sum_{i=1}^{P} \phi_i Z_{t-i} + \varepsilon_t + \sum_{j=1}^{Q} \theta_j \varepsilon_{t-j}$$

(Z_t: Differences; Z_{t-i}: Differenced; AR; MA)

ARIMA(1, 1, 1)

$$\Delta Y_t = c + \phi \Delta Y_{t-1} + \theta \cdot \varepsilon_{t-1} + \varepsilon_t$$

$$c = 2, \quad \phi = 0.6, \quad \theta = 0.4$$

$$\Delta Y_t = 10$$

$$\varepsilon_t = -3$$

$$\hat{\Delta Y}_{t+1} = 2 + 0.6 \times 10 + 0.4 \times (-3)$$

$$= 6.8$$

if the current level was 150

$$\hat{Y}_{t+1|t} = 150 + 6.8 = 156.8$$

prediction:

$$z_{t+1|t} = c + \phi_1 \cdot z_t$$

given that c and ϕ is learnt.

* SARIMA:

$$SARIMA(P,d,q)(P,D,Q)_m$$

$P \rightarrow$ seasonal AR order

$D \rightarrow$ seasonal differencing order

$Q \rightarrow$ seasonal MA order (lagged Error)

$m \rightarrow$ no. of observational in one season cycle

seasonal differencing:

D

$$\Delta_m Y_t = Y_t - Y_{t-m}$$

for weekly seasonal data: $m = 7$

$$\Delta_7 Y_t = Y_t - Y_{t-7}$$

How different is today from same weekday last week?

if $D = 0$ → no seasonal differencing

if $D = 1$ → subtract the previous seasonal value.

if $D = 2$ → seasonal difference twice

Adding both normal and seasonal differencing in SARIMA.

$$(1 - B)(1 - B^m) Y_t$$

AR → d SAR → D

$$W_t = Y_t - Y_{t-m}$$

$$Z_t = W_t - W_{t-1}$$

meaning of P:

Y_{t-m}, Y_{t-2m}, Y_{t-3m} - - -

$P = 2$, $m = 7$,	$P = 1$, $m = 7$
Y_{t-7}, Y_{t-14}	Y_{t-7}

meaning of Q.

In non-seasonal (q),

ε_{t-1}

In seasonal (Q) : ε_{t-m}

for our weekly data:

$$Y_t = C + \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

Put the SARIMA model together:

$$SARIMA(p,d,q)(P,D,Q)$$

$$\Phi(B)\,\Phi(B^m)\,(1-B)^d Y_t = C + \theta(B)\,\Theta(B^m)\,\varepsilon_t$$

- $\Phi(B)$ → ARIMA
- $\Phi(B^m)$ → seasonal
- $(1-B)^d$ → seasonal differencing
- $\theta(B)$ → AR → Arima
- $\Theta(B^m)$ (omega, P) → MA → seasonal

current behaviour = recent values + recent errors + seasonal values + seasonal errors + new random error

SARIMA (1, 1, 1)(1, 1, 1) →

$p = 1$ → use the prev. diff. value
$d = 1$ → use one ordinary diff.
$q = 1$ → use the prev. forecast Error

$P = 1$ → use the last seasonal diff. value.
$D = 1$ → use the last seasonal diff.
$Q = 1$ → use the prev. seasonal forecast Error.

$$RMSLE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left[\log(\hat{y}_i + 1) - \log(y_i + 1)\right]^2}$$

y	$\hat{y}$	Absolute Error	Absolute log error
10	20	10	$\log(21) - \log(11) \approx 0.28$
100	110	10	$\log(111) - \log(101) \approx 0.04$